

COMPLETE LEARNING SESSION

General-Mental-Ability-Integrated-Map - CSAT Learner-v2 Semantic Successor

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

LEARNING PATH



Deterministic fallback visual: move from precise concepts through argument and critique to exam application.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

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Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

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General-Mental-Ability-Integrated-Map - Complete CSAT Learning Session

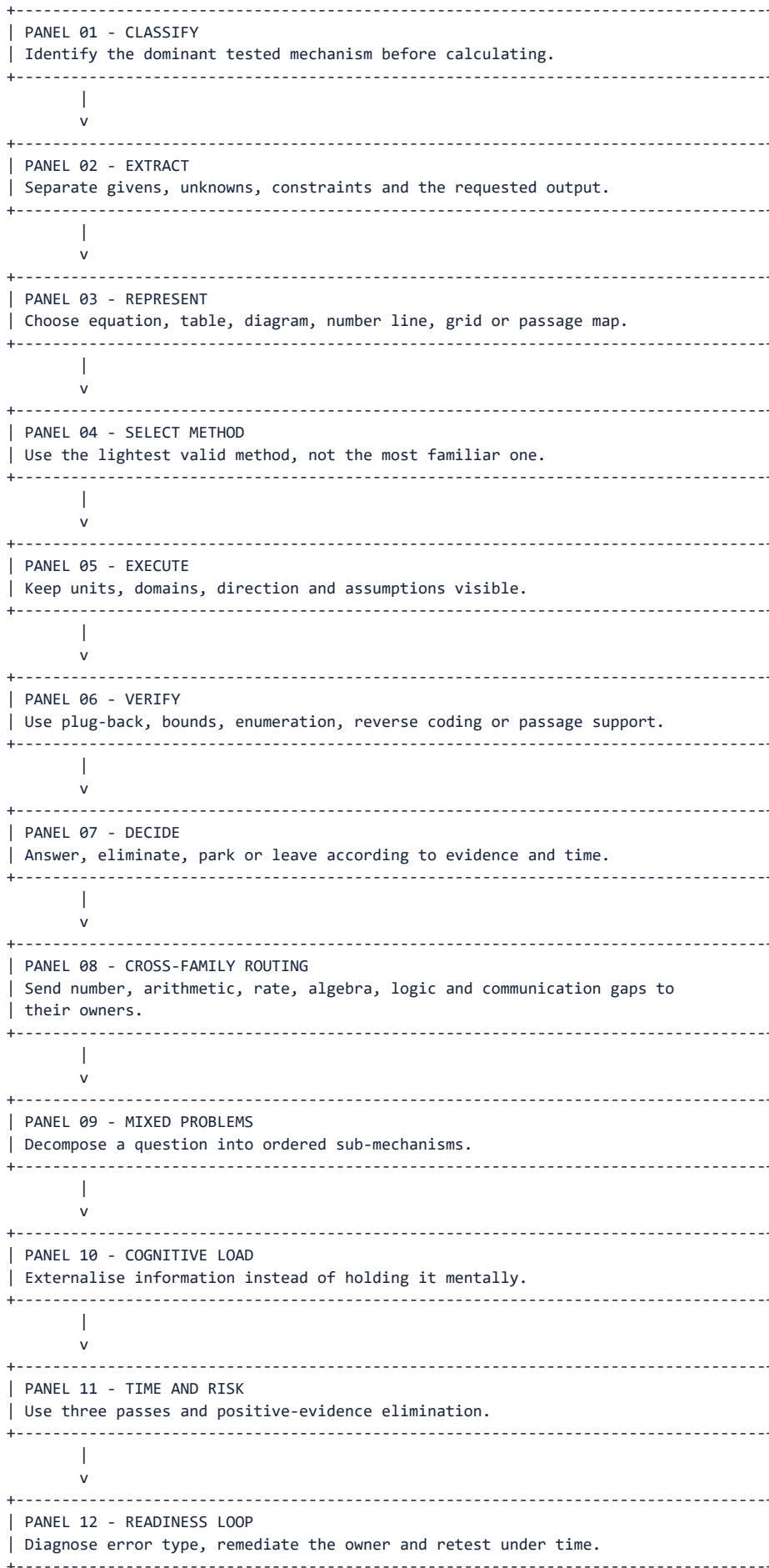
Identity: csat-08: learner-v2:g1 **Generation date:** 2026-09-06 **Approval:** false **Official syllabus anchor:** General mental ability.

Source	SHA-256 at generation
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The canonical Basic owner is taught first. Optional Advanced material is isolated after practice. All official-PYQ references preserve the repository's supplied/provisional key labels; no unavailable wording, official explanation or official key is invented.

BASIC LEARNING SESSION

Twelve-panel ASCII master flow



General Mental Ability - INTEGRATED CORE MAP

Subject: CSAT | **Tier:** Core integration spine | **Paper:** Prelims Paper II. **Official clause:** "General mental ability."
Purpose: Join the distributed number, arithmetic, rate, algebra, data and logical-reasoning owners into one complete decision map. Specialist formulas and drills remain in Topics 02-06. **Companion:**
 ../advanced/08_Mixed-General-Mental-Ability-and-Strategy.md

1. What General Mental Ability means in this architecture

General Mental Ability is an umbrella ability to:

UNDERSTAND information

- > REPRESENT it as number / relation / diagram / table / condition
- > SELECT a valid method
- > REASON or COMPUTE
- > CHECK sufficiency, consistency and magnitude
- > DECIDE under time and negative marking

It is broader than arithmetic and narrower than the whole CSAT paper. Reading comprehension has its own official clause and Topic 01, while communication has its own clause and Topic 07; both still support mental performance.

2. Complete ownership map

GMA capability	Primary Core owner	Typical forms
Number properties and pattern sense	Topic 02	Divisibility, remainders, digits, factors, HCF/LCM, powers, series
Quantitative proportional reasoning	Topic 03	Ratio, percentage, average, mixture, profit-loss, interest, ages
Rate, space and time reasoning	Topic 04	Work, pipes, speed, trains, boats, clocks, calendars, geometry
Algebraic and sufficiency judgement	Topic 05	Equations, inequalities, comparison, data sufficiency
Relational, spatial and symbolic reasoning	Topic 06	Arrangement, relations, directions, coding, deduction, counting, DI
Communication judgement	Topic 07	Barriers, feedback, conflict, channel, questionnaire, scenario choice

No question becomes ownerless merely because UPSC labels it broadly. Route it by the mechanism actually required.

3. First-read classification tree

Does the stem depend on a passage?

YES -> Topic 01

NO

|

- +-- asks about communication/conflict/channel? -> Topic 07
- +-- asks whether statements are sufficient? -> Topic 05
- +-- arrangement/relation/direction/code/table? -> Topic 06
- +-- divisibility/digits/remainder/pattern? -> Topic 02
- +-- ratio/%/average/money/age? -> Topic 03
- +-- work/speed/time/calendar/shape? -> Topic 04
- +-- equation/inequality/comparison? -> Topic 05

Some questions cross mechanisms. Select the dominant method first and use a second owner only when it contributes a distinct necessary step.

4. Universal GMA workflow

1. **Classify:** identify the mechanism, not the story.
2. **Extract:** write only the relevant quantities, conditions and relation words.
3. **Represent:** equation, ratio, table, set, grid, tree, compass or diagram.
4. **Choose least work:** property, elimination, estimation, substitution or full solution.
5. **Solve:** keep units and conditions visible.
6. **Verify:** magnitude, sign, parity, boundary, plug-back or counterexample.
7. **Decide:** mark, revisit or leave according to confidence and time.

Core rule: Externalise structure. Mental juggling creates avoidable working-memory errors.

Fully worked mixed-mechanism example

Question. Group A has 40 observations with mean 70. Group B has 60 observations with mean 40. Every Group B observation rises by 25%. Is the combined mean now above 55?

Classify -> extract -> represent: this is Topic 03 weighted-average reasoning plus a percentage change; it is **not** a simple-average question. Write totals, not a long list: A total = $40 \times 70 = 2800$; new B mean = $40 \times 1.25 = 50$; new B total = $60 \times 50 = 3000$.

Method selection and rejection: compare totals with the required threshold without dividing: $55 \times (40 + 60) = 5500$; actual total is 5800, so the combined mean is above 55. Reject $(70 + 50)/2 = 60$: that shortcut is unsafe because the groups have unequal sizes. Scaling B by 1.25 is safe only because the same percentage applies to **every** B observation.

Verification: $2800 + 3000 = 5800$ over 100 observations, so the exact mean is 58; it agrees with the inequality result.

Diagnosis: a 60 answer is a representation error (unweighted mean), while an incorrect B total is an execution error. Log the stage separately.

5. Cross-cutting reasoning distinctions

Distinction	Correct test
Necessary vs sufficient	In $A \rightarrow B$, A is sufficient and B is necessary
Implication vs converse	$A \rightarrow B$ permits not B $\rightarrow$ not A, not B $\rightarrow$ A
Possible vs certain	One valid case proves possibility; every valid case is needed for certainty
Correlation vs causation	Association alone does not establish the causal mechanism
Absolute vs relative change	Difference in units/points versus change relative to the original base
Average vs weighted average	Unequal groups require weights
Permutation vs combination	Order matters versus order does not
Distance vs displacement	Path length versus net positional change
Data sufficiency vs calculation	Prove uniqueness/sufficiency; do not solve more than required
Exact value vs estimation	Estimation is valid only when option separation makes it decisive

6. Representation toolkit

Information shape	Representation
Repeated proportional change	Multiplicative factors
Relative shares	Ratio or 100-unit base
Work/motion	Rate x time table
Family relation	Tree
Direction/turn	Compass sketch or coordinates
Seating/order	Slots or grid
Conditional statements	Arrows and contrapositives
Sets/overlap	Venn diagram or inclusion-exclusion
Data table/chart	Requested quantity, base and unit labelled first
Sufficiency	Separate Statement I, Statement II and combined tests

The representation is part of the solution, not decorative working.

7. Verification toolkit

- **Unit check:** kilometres cannot directly answer hours.
- **Magnitude check:** reject impossible orders of magnitude before exact calculation.
- **Parity/unit digit:** fast check for integer expressions.
- **Range/boundary:** probability must lie from 0 to 1; percentages and counts must fit the stem.
- **Plug-back:** substitute the proposed value into every condition.
- **Counterexample:** one valid counter-case defeats a claim of necessity.
- **Independent method:** use options or estimation to confirm a long calculation.
- **Completeness:** ensure every stated condition was used or deliberately shown irrelevant.

8. Operational time and attempt control

Stage	Hard operating rule
Classify	Spend 10-15 seconds identifying the owner and representation. If neither is visible, circle/park.
Pass 1 - Harvest	Spend at most 60 seconds on direct properties, one-step arithmetic, obvious diagrams and short tables. Mark only when the method is complete.
Pass 2 - Grind	Spend at most 120 seconds on multi-step quant, arrangements, complex DI and sufficiency. Write a compact representation before computing.
Park	At either cap, mark the question number, reason (method, representation, arithmetic, or live options) and move on. No emotional re-start.
Final review	Reserve the final 10 minutes for parked items, marked-option transcription and only elimination-backed decisions.

The 90-second paper average is not an individual-item command. Record whether a miss arose at classification, extraction, representation, execution, verification or decision; Topic 08 routes the repair, while the specialist topic supplies the drill.

9. Audited-PYQ integration rule

- The directly auditable question-level evidence in this repository is the 2024-2026 Set-A table in Question Audit Ledger. It should not be represented as an unsupported 2018-2023 routing database.
- 2024-2025 questions route to Topics 01-06 by their dominant mechanism. In 2026, Q72-Q77 retain Family 06 as an exclusive **structural** count but carry Topic 07 as their **content** owner.
- Topic 08 is the syllabus and navigation owner, not a duplicate omnibus table. Frequency, drills and remediation are read from the specialist owner and the tracker.

This avoids leaving an official umbrella ownerless, duplicating questions, or confusing solving format with syllabus content.

10. Readiness checklist

You are GMA-ready only if you can:

- classify a mixed question into the correct owner rapidly;
- translate words into a minimal representation;
- choose a property or shortcut only when its conditions hold;
- distinguish necessity, sufficiency, possibility and certainty;
- verify units, magnitude and conditions;
- abandon a time-expensive item without emotional attachment;
- diagnose the cause of each error and route revision to the correct topic.

Core firewall: Skipping Advanced Topic 08 cannot remove any GMA ownership, classification, workflow, representation, verification, time-control or revision-routing mechanism.

Semantic-completeness closure - 2026-09-06

Literal syllabus and canonical ownership

- **Literal clause:** General mental ability.
- **Canonical scope:** Owns the integrated classify-extract-represent-execute-verify-decide workflow, cross-family routing, cognitive-load control, time triage and readiness diagnostics.
- **Cross-topic boundary:** It does not duplicate formulas or full drills owned by Topics 02-07; it links them and teaches method selection for mixed questions.

Complete learner route

1. **Classify:** Identify the dominant tested mechanism before calculating.
2. **Extract:** Separate givens, unknowns, constraints and the requested output.
3. **Represent:** Choose equation, table, diagram, number line, grid or passage map.
4. **Select method:** Use the lightest valid method, not the most familiar one.
5. **Execute:** Keep units, domains, direction and assumptions visible.
6. **Verify:** Use plug-back, bounds, enumeration, reverse coding or passage support.
7. **Decide:** Answer, eliminate, park or leave according to evidence and time.
8. **Cross-family routing:** Send number, arithmetic, rate, algebra, logic and communication gaps to their owners.
9. **Mixed problems:** Decompose a question into ordered sub-mechanisms.
10. **Cognitive load:** Externalise information instead of holding it mentally.
11. **Time and risk:** Use three passes and positive-evidence elimination.
12. **Readiness loop:** Diagnose error type, remediate the owner and retest under time.

Verification and hostile-query gate

A mixed solution must name the owning mechanism, use a fitting representation, execute only justified steps and finish with an independent verification or risk decision.

The hostile absence search explicitly tested these families and close-option terms: **classify; extract; represent; execute; verify; decide; time control; error log**. A shortcut is usable only when its stated condition survives; otherwise return to the first-principles representation. Every worked answer must finish with an independent check: passage support, substitution, enumeration, units, bounds, reverse operation or option elimination.

Difficulty and timed progression

1. Foundation: recognise the family and state the governing definition or relation.
2. Core: solve a direct item with a visible representation and one verification.
3. Advanced: combine two mechanisms, test edge cases and reject close distractors.
4. Timed: use classify -> extract -> represent -> execute -> verify -> decide.
5. Remediation: log the error as concept, application, calculation, reading, passage, time or guess; return to the owning subtopic before retesting.

BASIC MCQS / REMEDIATION

Diagnostic and core set

Q1. Find the last digit of a large power.

Which canonical owner should be opened first?

- A. Topic 02 - Number Systems B. Topic 08 alone contains every formula C. No owner is relevant D. Choose by surface vocabulary only

Correct answer: A. Route by the dominant solving mechanism.

Q2. Find the last digit of a large power.

Which representation is most useful?

- A. Keep everything mentally B. Remainder/unit-digit cycle C. Write an unrelated essay D. Guess before classifying

Correct answer: B. Externalise the structure that directly encodes the givens.

Q3. Find the last digit of a large power.

Which verification should close the solution?

- A. Skip checking B. Use outside knowledge C. Check the cycle length D. Change the question

Correct answer: C. The check is independent of the main calculation or reasoning path.

Q4. Find the last digit of a large power.

What is the correct integrated sequence?

- A. Execute -> guess -> classify B. Read options -> assume -> stop C. Memorise -> ignore units -> mark D. Classify -> extract -> represent -> execute -> verify -> decide

Correct answer: D. The sequence prevents method error, hidden constraints and negative-value guesses.

Q5. Compare two discount offers.

Which canonical owner should be opened first?

A. Topic 03 - Arithmetic B. Topic 08 alone contains every formula C. No owner is relevant D. Choose by surface vocabulary only

Correct answer: A. Route by the dominant solving mechanism.

Q6. Compare two discount offers.

Which representation is most useful?

A. Keep everything mentally B. Percentage multipliers C. Write an unrelated essay D. Guess before classifying

Correct answer: B. Externalise the structure that directly encodes the givens.

Q7. Compare two discount offers.

Which verification should close the solution?

A. Skip checking B. Use outside knowledge C. Recompute from a common base D. Change the question

Correct answer: C. The check is independent of the main calculation or reasoning path.

Q8. Compare two discount offers.

What is the correct integrated sequence?

A. Execute -> guess -> classify B. Read options -> assume -> stop C. Memorise -> ignore units -> mark D. Classify -> extract -> represent -> execute -> verify -> decide

Correct answer: D. The sequence prevents method error, hidden constraints and negative-value guesses.

Q9. Two workers complete a job together.

Which canonical owner should be opened first?

A. Topic 04 - Rates B. Topic 08 alone contains every formula C. No owner is relevant D. Choose by surface vocabulary only

Correct answer: A. Route by the dominant solving mechanism.

Q10. Two workers complete a job together.

Which representation is most useful?

A. Keep everything mentally B. Work-rate table C. Write an unrelated essay D. Guess before classifying

Correct answer: B. Externalise the structure that directly encodes the givens.

Q11. Two workers complete a job together.

Which verification should close the solution?

A. Skip checking B. Use outside knowledge C. Add rates, not days D. Change the question

Correct answer: C. The check is independent of the main calculation or reasoning path.

Q12. Two workers complete a job together.

What is the correct integrated sequence?

A. Execute -> guess -> classify B. Read options -> assume -> stop C. Memorise -> ignore units -> mark D. Classify -> extract -> represent -> execute -> verify -> decide

Correct answer: D. The sequence prevents method error, hidden constraints and negative-value guesses.

Q13. Decide whether two statements uniquely fix x.

Which canonical owner should be opened first?

A. Topic 05 - Data Sufficiency B. Topic 08 alone contains every formula C. No owner is relevant D. Choose by surface vocabulary only

Correct answer: A. Route by the dominant solving mechanism.

Q14. Decide whether two statements uniquely fix x.

Which representation is most useful?

A. Keep everything mentally B. Statement-by-statement case test C. Write an unrelated essay D. Guess before classifying

Correct answer: B. Externalise the structure that directly encodes the givens.

Q15. Decide whether two statements uniquely fix x.

Which verification should close the solution?

A. Skip checking B. Use outside knowledge C. Search for a second valid value D. Change the question

Correct answer: C. The check is independent of the main calculation or reasoning path.

Q16. Decide whether two statements uniquely fix x.

What is the correct integrated sequence?

A. Execute -> guess -> classify B. Read options -> assume -> stop C. Memorise -> ignore units -> mark D. Classify -> extract -> represent -> execute -> verify -> decide

Correct answer: D. The sequence prevents method error, hidden constraints and negative-value guesses.

Remediation protocol

1. Recompute without options.
2. Name the failed rule or passage phrase.
3. Reject every distractor for a specific reason.
4. Retry with altered numbers, wording or constraints.
5. Advance only after two consecutive correct answers under the time ceiling.

PYQS AND ANSWER PRACTICE

Verified 2024-2026 Set-A demand and key ledger

Year	Q	Neutral verified demand	Key status	Solution architecture
-	-	No direct question is duplicated here; use the cross-owner routes.	-	Route to Topics 01-07 by mechanism.

The table is a non-verbatim routing audit. It records the locally checked Set-A key letter and a valid solving route, but does not pretend that UPSC publishes model solutions. The separate workbook supplies original solved equivalents for every mechanism.

Timed mixed transfer

Q33. A train crosses a platform.

Which canonical owner should be opened first?

A. Topic 04 - Motion B. Topic 08 alone contains every formula C. No owner is relevant D. Choose by surface vocabulary only

Correct answer: A. Route by the dominant solving mechanism.

Q34. A train crosses a platform.

Which representation is most useful?

A. Keep everything mentally B. Distance-speed-time equation C. Write an unrelated essay D. Guess before classifying

Correct answer: B. Externalise the structure that directly encodes the givens.

Q35. A train crosses a platform.

Which verification should close the solution?

A. Skip checking B. Use outside knowledge C. Include both lengths D. Change the question

Correct answer: C. The check is independent of the main calculation or reasoning path.

Q36. A train crosses a platform.

What is the correct integrated sequence?

A. Execute -> guess -> classify B. Read options -> assume -> stop C. Memorise -> ignore units -> mark D. Classify -> extract -> represent -> execute -> verify -> decide

Correct answer: D. The sequence prevents method error, hidden constraints and negative-value guesses.

Q37. A number leaves specified remainders.

Which canonical owner should be opened first?

A. Topic 02 - Number Systems B. Topic 08 alone contains every formula C. No owner is relevant D. Choose by surface vocabulary only

Correct answer: A. Route by the dominant solving mechanism.

Q38. A number leaves specified remainders.

Which representation is most useful?

A. Keep everything mentally B. Congruence table C. Write an unrelated essay D. Guess before classifying

Correct answer: B. Externalise the structure that directly encodes the givens.

Q39. A number leaves specified remainders.

Which verification should close the solution?

A. Skip checking B. Use outside knowledge C. Plug back into every modulus D. Change the question

Correct answer: C. The check is independent of the main calculation or reasoning path.

Q40. A number leaves specified remainders.

What is the correct integrated sequence?

A. Execute -> guess -> classify B. Read options -> assume -> stop C. Memorise -> ignore units -> mark D. Classify -> extract -> represent -> execute -> verify -> decide

Correct answer: D. The sequence prevents method error, hidden constraints and negative-value guesses.

OPTIONAL ADVANCED DEPTH - NOT REQUIRED FOR A CORE ANSWER

Mixed General Mental Ability and Strategy - OPTIONAL ADVANCED

Subject: CSAT | **Tier:** Optional Advanced. **Prerequisite:**

../basic/08_General-Mental-Ability-Integrated-Map.md and the relevant specialist Core files, Topics 02-07. **Firewall:** No indispensable formula, distinction or PYQ is owned only here.

1. Mixed-mechanism problems

Hard GMA questions often combine:

classification + representation + one quantitative/logical mechanism + verification

Examples:

- a seating condition followed by a counting question;
- a rate problem with a ratio change;
- a data table requiring weighted average and percentage-point distinction;
- a conditional chain followed by a sufficiency judgement.

Do not solve all visible mathematics. Identify the exact output the stem requires.

2. Method-selection hierarchy

Try methods in this order:

1. direct invariant/property;
2. option elimination;
3. smart values or scaling;
4. compact algebra/diagram;
5. full enumeration or long computation.

A shortcut is safe only when it preserves every condition and the option spacing makes the result decisive.

3. Cognitive-load control

- chunk repeated information into a table or symbol;
- write intermediate results rather than rehearsing them mentally;
- standardise diagrams for relations, directions and rates;
- pause after the setup to check the question asked;
- separate solving from option marking to reduce transcription errors.

Working memory is limited; external notation is a performance tool.

4. Robustness under uncertainty

State	Response
Method known, arithmetic long	Estimate or inspect options before calculating fully
Two options survive	Test the condition that distinguishes them
Stem appears inconsistent	Re-read qualifiers, units and shared directions
Diagram has branches	Seek a forced conclusion or counterexample rather than enumerate blindly
Confidence remains low	Park the item and protect time

5. Advanced error analytics

Track errors by **mechanism and process stage**:

Stage	Failure	Remediation
Classification	Chose wrong topic/method	Timed mixed-identification drills
Extraction	Missed condition or qualifier	Underline quantities and logical words
Representation	Wrong equation/grid/tree	Rebuild from the stem before calculating
Execution	Arithmetic or logical slip	Smaller steps and intermediate checks
Verification	Did not test result	Mandatory plug-back/magnitude habit
Decision	Over-attempted or abandoned too early	Confidence and time log

Raw score alone cannot reveal which stage needs repair.

6. Mixed-session design

A useful advanced session contains:

- rapid classification without solving;
- one representative problem from each weak mechanism;
- a timed mixed block;
- error reconstruction without looking at the solution;
- a second attempt after a delay.

Advanced practice should increase transfer and decision quality, not merely increase difficulty.

Timed advanced mini-drill protocol

Use a 10-item mixed block: **10-15 seconds** to classify each item without solving, then a second pass of **60 seconds** for direct items and **120 seconds** for selected hard items. Write the stage failure beside every miss: classification, extraction, representation, execution, verification or decision. Park any item at its cap and keep the last **10 minutes of a full paper** for review; these limits are operating discipline, not an official UPSC timing rule.

Optional worked transfer check

For the Core Topic 08 weighted-average example, an advanced solver first compares totals against 55×100 ; no division or decimal conversion is needed. The shortcut is valid only after preserving the group sizes and applying the 25% factor uniformly to Group B. A solver who averages 70 and 50 has selected the wrong representation, not merely made an arithmetic slip.

CONSOLIDATED REGISTER NOTES

Complete revision spine

- **Classify:** Identify the dominant tested mechanism before calculating.
- **Extract:** Separate givens, unknowns, constraints and the requested output.
- **Represent:** Choose equation, table, diagram, number line, grid or passage map.
- **Select method:** Use the lightest valid method, not the most familiar one.
- **Execute:** Keep units, domains, direction and assumptions visible.
- **Verify:** Use plug-back, bounds, enumeration, reverse coding or passage support.
- **Decide:** Answer, eliminate, park or leave according to evidence and time.
- **Cross-family routing:** Send number, arithmetic, rate, algebra, logic and communication gaps to their owners.
- **Mixed problems:** Decompose a question into ordered sub-mechanisms.
- **Cognitive load:** Externalise information instead of holding it mentally.
- **Time and risk:** Use three passes and positive-evidence elimination.
- **Readiness loop:** Diagnose error type, remediate the owner and retest under time.

Ownership and close-option firewall

- **Own here:** Owns the integrated classify-extract-represent-execute-verify-decide workflow, cross-family routing, cognitive-load control, time triage and readiness diagnostics.
- **Do not duplicate:** It does not duplicate formulas or full drills owned by Topics 02-07; it links them and teaches method selection for mixed questions.
- **Verification:** A mixed solution must name the owning mechanism, use a fitting representation, execute only justified steps and finish with an independent verification or risk decision.

Timed answer route

CLASSIFY -> EXTRACT -> REPRESENT -> EXECUTE -> VERIFY -> DECIDE

- Use estimation or option elimination only after preserving the governing condition.
- A blank costs zero; a rushed unsupported answer also consumes time and may attract negative marks.
- For every error, record concept/application/calculation/reading/passage/time/guess, repair the owner, and retry a new item rather than memorising the old option.